Linear Programming

1 Foundations of Linear Programming

What is linear programming (LP)? It is a **mathematical optimization framework** for making decisions under constraints.

1.1 Formulating Linear Programming Problems

Kaiser and Messer emphasize that formulation is the central task in LP: the analyst translates a verbal decision problem into a mathematical model. Begin by reading the problem carefully to understand the decision setting before trying to write equations. Then read it again to sort the information needed for the model.

1.1.1 A Practical Formulation Sequence

1. **Identify the objective.** What is the decision maker trying to maximize or minimize—for example, profit, cost, sales, or production?
2. **Identify the activities (decision variables).** What choices can the decision maker make to pursue the objective? Define the unit of measurement for each activity.
3. **Identify objective coefficients.** For each activity, determine its per-unit contribution to the objective and record its units.
4. **Identify resources or requirements.** List the resources controlled by the decision maker, their available amounts, and their units. In problems without fixed resources, identify the relevant right-hand-side requirements instead.
5. **Identify technical coefficients.** For every activity-resource pair, determine how much of the resource is used or supplied by one unit of the activity. Check the units carefully.
6. **Write the structural constraints.** Translate resource limits or minimum requirements into $\leq$, $\geq$, or $=$ constraints.
7. **Add variable restrictions and assemble the model.** Include nonnegativity (or other sign restrictions) and combine the objective and constraints into a complete scalar model. Descriptive constraint labels make the model easier to audit and interpret.

Throughout the process, check that every term within an equation has compatible units. Formulation improves through practice: solve many verbal problems, compare alternative specifications, and ask whether each equation faithfully represents the decision setting.

1.2 Farm LP Example

A farmer has 500 acres of land available and is deciding how to allocate it between wheat and corn. Each acre of wheat yields a profit of \$200, requires 3 hours of labor, and 4 units of fertilizer. Each acre of corn yields a profit of \$300, requires 4 hours of labor, and 3 units of fertilizer. The farm has at most 1,800 hours of labor available and 2,000 units of fertilizer.

Formulate this situation as a linear programming problem. Clearly define the decision variables, write down the objective function representing total profit, and specify the constraints that capture the land, labor, fertilizer, and nonnegativity restrictions.

Decision variables.
Let W = acres of wheat, C = acres of corn.

1.2.1 Scalar form

$$\begin{aligned}
\max_{W,C} \quad & 200\,W + 300\,C \\
\text{s.t.} \quad & W + C \le 500 && \text{(land)} \\
& 3W + 4C \le 1800 && \text{(labor)} \\
& 4W + 3C \le 2000 && \text{(fertilizer)} \\
& W,\ C \ge 0\ .
\end{aligned}$$

1.2.2 Matrix (compact) form

$$\begin{aligned}
\max_{x\in\mathbb{R}^2_{\ge 0}} \quad & c^\top x \\
\text{s.t.} \quad & Ax \le b,
\end{aligned}
\qquad \text{with} \quad x = \begin{bmatrix} W \\ C \end{bmatrix},\ c = \begin{bmatrix} 200 \\ 300 \end{bmatrix},\ A = \begin{bmatrix} 1 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix},\ b = \begin{bmatrix} 500 \\ 1800 \\ 2000 \end{bmatrix}.$$

1.3 Properties of LP

- **Objective** is to either maximize or minimize
- **Constraints**: linear inequalities or equalities that limit the feasible choices (e.g., land, labor, fertilizer).
- All equations are **linear**
- **Decision variables**: the choices to be made (e.g., how many acres of wheat and corn to plant) are non-negative.

1.4 Assumptions of Linear Programming

The first three assumptions concern whether the model represents the decision problem appropriately; the remaining four state its mathematical and information requirements.

- **Objective-function appropriateness:** one stated objective is sufficient to rank feasible choices.
- **Decision-variable appropriateness:** all relevant choices are included and controlled by the decision maker.
- **Constraint appropriateness:** the constraints accurately represent the relevant limits, requirements, and units.
- **Proportionality:** each activity's per-unit contribution and resource use do not change with its level.
- **Additivity:** the contribution of one activity does not depend on the level of another activity.
- **Divisibility:** decision variables may take fractional values.
- **Certainty:** objective, technical, and right-hand-side coefficients are known fixed values.

When an assumption is unsuitable, the model may need an extension: multi-objective programming, integer programming, nonlinear programming, or stochastic programming.

1.5 Graphical Solution Method (Two Variables)

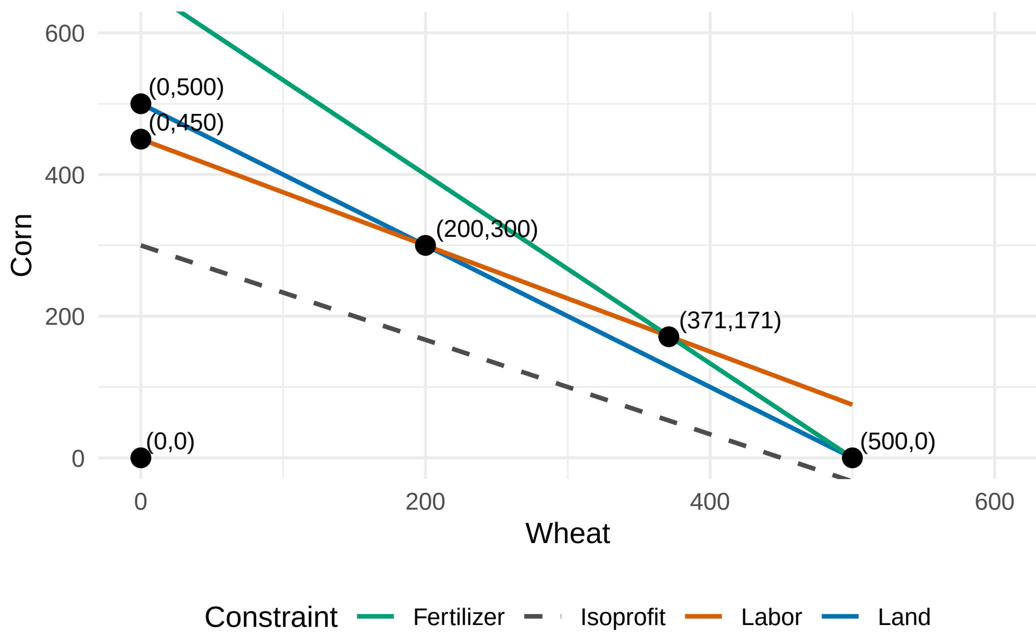
1.5.1 Key Geometry Terms

- A **feasible solution** satisfies every constraint; an **infeasible solution** violates at least one.
- The **feasible region** is the set of all feasible solutions.
- The feasible region of an LP is **convex**: the line segment joining any two feasible solutions is also feasible.
- An **extreme point** (or vertex) is a corner of the feasible region.
- A constraint is **binding** when it holds with equality at a solution; otherwise, it has **slack**.

Step 1. Draw the constraints.

- Land: $W + C \leq 500$
- Labor: $3W + 4C \leq 1800$
- Fertilizer: $4W + 3C \leq 2000$
- Nonnegativity: $W, C \geq 0$
- Plot based on endpoints. Set one variable to zero and devote all resources to the other variable.

```
Warning: package 'ggplot2' was built under R version 4.5.2
```



Step 2. Identify the feasible region.

- Intersection of all constraints in the (W, C) plane.
- Polygon bounded by lines.

Step 3. Plot the objective function.

- Profit $= 200W + 300C$.
- Show isoprofit lines:
  - Suppose we plot \$90,000 profit. All wheat no corn, then all corn no wheat.
  - Lines of constant profit have slope $-\frac{200}{300} = -\frac{2}{3}$.

Step 4. Locate the optimum.

- Slide the isoprofit line outward until the last point of contact with the feasible region.
- Optimum is always at a **corner point** (fundamental theorem of LP).

Step 5. Verify algebraically.

Use the binding constraints to recover the exact solution, then check every constraint and calculate the objective value. In the farm example, the graphical optimum has $W = 0$ and the labor constraint binds:

$$4C = 1800 \quad \Rightarrow \quad C = 450.$$

The solution $(W, C) = (0, 450)$ satisfies land use $(450 \leq 500)$ and fertilizer use $(1350 \leq 2000)$, and it yields profit $200(0) + 300(450) = 135{,}000$.

1.6 Section 1 Summary

- LP is a framework for making decisions subject to constraints.
- Formulate an LP by identifying the objective, activities and their units, coefficients, resources or requirements, and variable restrictions.
- A standard LP combines a linear objective, linear structural constraints, and sign restrictions; its coefficients are summarized in c, A, and b.
- LP relies on assumptions about appropriate objectives, variables, and constraints, along with proportionality, additivity, divisibility, and certainty.
- In two-variable problems, graph the constraints to find the feasible region, use an isoprofit line to locate an extreme-point optimum, and verify the result algebraically.

2 Solving Linear Programs

2.1 Simplex Method

2.1.1 Why We Need It

- The graphical method only works for **two variables**.
- Real problems may involve **hundreds or thousands** of variables.
- Key geometric fact:
  - The feasible region of an LP is a **convex polytope**.

  - The **optimal solution lies at a vertex (corner point)**. What about the problem makes this a fact?
- The simplex method provides a **systematic way** to move from vertex to vertex until the best one is found.

2.1.2 Core Idea

- Start at a **basic feasible solution (BFS)** — a corner point of the feasible region.
- At each step:
  1. Identify an **entering variable**: an activity whose introduction can improve the objective.
  2. Determine which current activity must decrease to keep the solution feasible; this is the **leaving variable**.
  3. **Pivot** to a neighboring BFS.
- Stop when no variable can improve the objective — this is optimal.

2.1.3 Intuition

- Simplex is like **walking along the edges** of the feasible polygon.
- At each corner, ask: *"If I move along this edge, does profit go up?"*
- Continue until no edge yields improvement.
- The same logic applies in higher dimensions, even though we cannot draw the polytope.

2.1.4 What to Take from Simplex

- A **basis** identifies the activities currently used to describe a corner-point solution.
- **Reduced costs** summarize whether introducing a currently inactive activity can improve the objective.
- The **ratio test** preserves feasibility when the algorithm moves to the next corner.
- A **slack variable** records unused capacity in a "$\leq$" constraint. A positive slack means that the corresponding resource is not fully used.

Modern solvers carry out the tableau and matrix calculations. In this course, the emphasis is on formulating the problem, recognizing the logic of a pivot, and interpreting the solution—not hand-executing a long simplex tableau.

2.1.5 Optional Reference for Simplex Mechanics

For a matrix-algebra treatment of the simplex algorithm, including tableau mechanics, initial feasible bases, and Phase I/Phase II methods, see McCarl and Spreen, Chapter III, especially Sections 3.2 and 3.4. The graphical method above supplies the intuition needed to interpret what the algorithm is doing.

2.1.6 Excel Solver

Excel implements the simplex method in the solver add-on. See LP_land_alloc.xlxs

2.2 Day 2 Summary

- Linear programming problems have:
  - **Decision variables** (choices to make),
  - **Objective function** (profit, cost, etc.),
  - **Constraints** (resource limits, requirements).
- Graphical method (2 variables) shows:
  - Feasible region = convex polygon.
  - Optimum occurs at a **corner point**.
- **Fundamental theorem of LP:** optimum is always at a vertex of the feasible region.
- **Simplex method** generalizes:
  - Moves systematically from one **basic feasible solution (BFS)** to another.
  - Uses **entering and leaving variables** to pivot.
  - Stops when no further improvement is possible.
- **Slack variables** convert inequalities to equalities and measure unused resources.
- Solvers carry out the pivot calculations; our focus is their modeling and economic interpretation.

3 Sensitivity Analysis

3.1 Review

- LPs make 7 assumptions. What are a few?
- How do LPs algorithms solve? What is the fundamental theorem of LP?

3.2 Motivation

- LPs give an **optimal solution** and an **objective value**.
- But: parameters (profits, resources, input requirements) are rarely known with certainty.
- We need to know how **robust** the solution is.

3.3 What Is Sensitivity Analysis?

- Also called **post-optimality analysis**.
- Asks: *how much can model parameters change before the current solution changes?*
- Focus on three categories:
  1. RHS (resources)

  2. Objective coefficients (profits/costs)

  3. Technical coefficients (input-output relationships)

3.4 RHS (Right-Hand Side) Ranging

- Suppose resource availability changes:

$$b_{new} = b_{old} + \Delta r$$

- Current solution remains optimal **as long as basic variables stay $\geq$ 0**.
- Interpretation:
  - Within the allowable range, the **shadow price** is valid.
  - Outside the range, the optimal activity mix shifts.

3.5 Objective Coefficient Ranging

- How much can a profit coefficient change before the basis changes?
- For **nonbasic variables**: reduced costs must remain ≥ 0.
- For **basic variables**: check feasibility of objective row with new coefficient.

3.6 Technical Coefficient Changes

- What if technology or input requirements change?
- Example: wheat now needs 3.5 instead of 3 labor hours per acre.
- Sensitivity analysis uses shadow prices to approximate the effect on objective value.
- Interpretation: tighter labor constraints may shift which crop dominates.

3.7 The 100% Rule

The 100% Rule provides a way to evaluate whether simultaneous changes in objective function coefficients or right-hand side (RHS) values will preserve the current optimal basis — that is, whether the current solution remains optimal without re-solving the problem.

This rule applies separately to:

- Changes in objective function coefficients (i.e., the c_i's), and
- Changes in the RHS values of constraints (i.e., the b_j's).

Objective Function Coefficients

Suppose multiple objective coefficients change. For each decision variable x_i, let:

- Δc_i be the change in the objective coefficient.
- AL_i, AU_i be the allowable decrease and increase, respectively, as reported in the sensitivity analysis.

Define the proportion of the allowable range used for each change:

$$r_i = \begin{cases} \frac{|\Delta c_i|}{AL_i} & \text{if } \Delta c_i < 0 \\ \frac{|\Delta c_i|}{AU_i} & \text{if } \Delta c_i > 0 \end{cases}$$

Then, compute: $\sum_{i \in \text{changed vars}} r_i$. If: $\sum r_i \leq 1$, then the current optimal basis remains optimal, although the optimal value of the objective function will generally change. If the sum exceeds 1, the basis may change, and re-solving the problem is required.

3.8 Wheat–Corn Example (Excel Solver Report)

3.8.1 Excel Top Panel: Variable Cells

Interpretation of columns:

- **Final Value** – the optimal level of the decision variable (e.g., acres of wheat or corn).
- **Reduced Cost** – for nonbasic variables (value = 0), how much the objective coefficient must improve before the variable would enter the solution. Zero if the variable is positive in the solution.
- **Objective Coefficient** – the profit (or cost) per unit used in the objective function.
- **Allowable Increase / Decrease** – the range over which the objective coefficient can change without altering the current optimal basis (solution structure).

Interpretation of results

- Corn: optimal solution plants 450 acres of corn. Reduced cost = 0 because it's in the basis.
- Wheat: optimal solution plants 0 acres of wheat. Reduced cost = -25 means if wheat's profit increased by more than $25/acre (from 200 → 225), it would enter the solution.

Ranges:

- Wheat: profit can increase up to +25 before wheat enters.
- Corn: profit can fall as much as 33.3 (300 → 266.7) before solution changes.
- Interpretation: Corn dominates under current prices. Wheat only becomes attractive if its relative profit improves substantially.

3.8.2 Excel Bottom Panel: Constraints

Interpretation of columns:

- **Final Value** – the amount of the resource actually used at the solution.
- **Constraint R.H. Side** – the available amount of the resource (the right-hand side of the inequality).
- **Shadow Price** – the marginal value of relaxing the constraint (increase in objective if RHS increases by 1 unit), valid only within the allowable range.
- **Allowable Increase / Decrease** – the range over which the shadow price remains valid and the current basis stays optimal.

Interpretation of results:

- Land: only 450 acres used out of 500. Slack = 50 acres. Shadow price = 0 because land is not binding. You can increase land indefinitely without improving profit (since labor is the true bottleneck).
- Labor: fully used (1800/1800). Shadow price = 75 means each additional labor hour would increase profit by $75, valid for up to +200 extra hours.
- Fertilizer: only 1350 units used of 2000. Slack = 650. Shadow price = 0 because it's not binding.

Economic Takeaways for Discussion

- Which resource is scarce? Labor.
- How much should the agent be willing to pay for an additional unit of labor? 75 per labor hour. This is also the opportunity cost of labor on this operation.
- Why does wheat drop out of the solution? Because relative to corn it uses too much labor per profit dollar.
- Policy thought experiment: If labor availability were increased by 200 hours, profit would rise by $15,000 (200 × 75).

3.9 Economic Interpretation

- **Shadow prices**: marginal value of resources.
- **Allowable ranges**: robustness of those shadow prices.
- **Managerial use**:
  - Identify which resources are most binding.
  - Assess which profit coefficients are critical.
  - Evaluate new technologies or policy changes.

3.10 Summary

- Sensitivity analysis extends LP results beyond a single point solution.

- Provides insight into:
  - Resource valuation (RHS changes),
  - Profit robustness (objective changes),
  - Technology shifts (coefficient changes).

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4 Getting Started with R & RStudio

4.1 RStudio Orientation (2–3 Minute Tour)

- **Source Pane (top-left):** where you edit scripts (`.R`) and notebooks (`.qmd`, `.Rmd`).
- **Console (bottom-left):** runs commands immediately (`>` prompt).
- **Environment/History (top-right):** objects in memory (data, vectors, functions).
- **Files/Plots/Packages/Help (bottom-right):** manage files, see plots, install packages, read docs.

Workflow tips

- Create a **Project** (`File → New Project…`) for the course; it pins your working directory.
- Put scripts in a `code/` folder, data in `data/`, and outputs in `out/`.
- Use **scripts** for anything you might need to re-run. Avoid one-off console work for assignments.

4.2 Preview

- Next class: **duality** — formalize the relationship between primal and dual problems.
- Show how shadow prices emerge naturally from the dual formulation.

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5 Duality in Linear Programming

5.1 Review

- Sensitivity analysis evaluates robustness of LP solutions to parameter changes.
- Shadow prices measure the marginal value of resources.
- Ranging analysis indicate how much parameters can change before the solution structure shifts.
- The 100% Rule helps assess simultaneous changes in parameters.
- Excel Solver provides a convenient way to solve LPs and get sensitivity reports.

5.2 Motivation for Duality

- Duality provides a **theoretical foundation** for shadow prices.
- Every LP (the **primal**) has a corresponding **dual** LP.
- Solutions to the dual give the **shadow prices** of the primal constraints.
- Duality reveals deep connections between resources and values.
- Duality also aids in sensitivity analysis and economic interpretation.

5.3 Canonical primal–dual pair

Primal (max form, $\leq$ constraints):

$$\begin{aligned} \max_{x \geq 0} \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

Dual (min form, $\geq$ constraints):

$$\begin{aligned} \min_{y \geq 0} \quad & b^\top y \\ \text{s.t.} \quad & A^\top y \geq c \end{aligned}$$

Rules of transformation (quick guide):

- One **dual variable** per **primal constraint**; one **dual constraint** per **primal variable**.
- Primal "$a_i^T x \leq b_i$" with $x \geq 0 \rightarrow$ dual variable $y_i \geq 0$. Relaxing the RHS makes the feasible set bigger, so the dual variable (shadow price) must be nonnegative.
- Primal "$a_i^T x \geq b_i$" $\rightarrow$ dual variable $y_i \leq 0$. Tightening the RHS makes the feasible set bigger (since it's "$\geq$"), so the associated price flips sign.
- Primal "$a_i^T x = b_i$" $\rightarrow$ dual variable y **free** (unrestricted in sign). An equality can cut the feasible set in either direction, so the shadow price can be positive or negative.
- If a primal variable is **free**, the corresponding dual constraint is an **equality**; if primal variable has sign $x_j \leq 0$, flip inequality accordingly.

5.4 Wheat–Corn primal and its dual

Primal (wheat–corn):

$$\begin{aligned} \max\ & 200\,x_W + 300\,x_C \\ \text{s.t.}\ & x_W + x_C \leq 500 && \text{(land)} \\ & 3x_W + 4x_C \leq 1800 && \text{(labor)} \\ & 4x_W + 3x_C \leq 2000 && \text{(fertilizer)} \\ & x_W,\ x_C \geq 0\ . \end{aligned}$$

Introduce dual variables $y_1, y_2, y_3 \geq 0$ for (land, labor, fertilizer).

Dual:

$$\begin{aligned} \min\ & 500\,y_1 + 1800\,y_2 + 2000\,y_3 \\ \text{s.t.}\ & y_1 + 3y_2 + 4y_3 \geq 200 && \text{(wheat)} \\ & y_1 + 4y_2 + 3y_3 \geq 300 && \text{(corn)} \\ & y_1, y_2, y_3 \geq 0\ . \end{aligned}$$

Economic meanings:

y_1= land rent (per acre), y_2= wage (per labor hour), y_3= fertilizer value (per unit).

5.4.1 Solve directly by intuition

A low-cost feasible choice is to try just one variable:

- Set $y_1 = 0$, $y_3 = 0$. The corn constraint gives $4y_2 \geq 300 \Rightarrow y_2 \geq 75$.
- Check wheat: $3y_2 \geq 200 \Rightarrow y_2 \geq 66.67$. So $y_2 = 75$ satisfies both.
- Dual objective: $500(0) + 1800(75) + 2000(0) = 135{,}000$.
- Could adding anything to y_1 or y_3 lower the cost?
- No, because both constraints are already satisfied and any increase would only raise the objective.
- Feasibility check:
  - Wheat: $0 + 3(75) + 4(0) = 225 \geq 200$
  - Corn: $0 + 4(75) + 3(0) = 300 \geq 300$

Let's verify in excel

5.5 Duality theorems

5.5.1 Weak duality

For any feasible solution x and y,

$$c^\top x \leq b^\top y$$

- From the primal constraints: $Ax \leq b$
- From the dual constraints: $A^\top y \geq c$
- Premultiply the dual constraint by $x^\top \geq 0$: $x^\top A^\top y \geq x^\top c$ and note that $x^\top A^\top y = (Ax)^\top y$
- And post multiply the primal constraint by y: $(Ax)^\top y \leq b^\top y$.
- Then putting the two together gives $c^\top x \leq x^\top A^\top y = (Ax)^\top y \leq b^\top y$.

Wheat-corn example:

- Primal feasible: $x_W = 0, x_C = 450 \Rightarrow Z = 135{,}000$.
- Dual feasible: $y_1 = 0, y_2 = 75, y_3 = 0 \Rightarrow W = 500(0) + 1800(75) + 2000(0) = 135{,}000$.
- Weak duality holds: $135{,}000 \leq 135{,}000$.

5.5.2 Strong duality

If the primal and dual have optimal solutions, so does the other, and $c^\top x^* = b^\top y^*$.

- At the optimum, the primal objective (profit) equals the dual objective (resource value).
- The upper bound becomes tight: best possible activity plan = best possible valuation of resources.
- Market analogy:
  - Primal: maximize profit given resource costs.
  - Dual: minimize resource costs to support given profits.
  - At equilibrium, profit = cost of resources.
- Wheat–corn example:
  - Primal optimum: $Z^* = 135{,}000$.
  - Dual optimum: $500(0) + 1800(75) + 2000(0) = 135{,}000$

5.5.3 Complementary slackness conditions

The **complementary slackness conditions** link the primal and dual solutions. They provide the bridge between activity levels and resource prices:

- For each **primal constraint** $a_i^\top x \leq b_i$ with dual variable $y_i \geq 0$:

  $$y_i\,(b_i - a_i^\top x) = 0$$

  This means that either:
  - The constraint is **binding** ($a_i^\top x = b_i$) and then $y_i \geq 0$,

  - Or the constraint is **slack** ($a_i^\top x < b_i$) and then $y_i = 0$.
- For each **primal variable** $x_j \geq 0$ with dual inequality $a_j^\top y \geq c_j$:

  $$x_j\,(a_j^\top y - c_j) = 0$$

  This means that either:
  - The activity is **produced** ($x_j > 0$) and then its dual inequality binds exactly ($a_j^\top y = c_j$),

  - Or the activity is **not produced** ($x_j = 0$) and then the dual inequality can be slack ($a_j^\top y > c_j$).

5.6 Summary

- Every LP has a corresponding dual LP.
- Dual variables represent shadow prices of primal constraints.
- Weak duality: primal objective $\leq$ dual objective for any feasible solutions.
- Strong duality: at optimality, primal objective = dual objective.
- Complementary slackness links primal and dual solutions, indicating which constraints and variables are binding.
- Duality provides a theoretical foundation for sensitivity analysis and economic interpretation of LP results.

6 Extension: Input–Output Modeling

This final section is an extension of the core LP material. Input–output analysis is primarily a **linear accounting model** for tracing inter-industry linkages; its connection to LP and duality is developed at the end of the section.

6.1 Motivation

- Economies are networks of interdependent sectors.
- The output of one sector (e.g., steel) may be an input for another (e.g., car manufacturing).
- Understanding these linkages helps us analyze how a change in demand or policy ripples through the entire economy.
- Input–output (I–O) analysis was developed by **Wassily Leontief** (Nobel Prize, 1973).

6.2 Basic Input–Output Structure

- Consider an economy with n sectors.
- **Notation:**
  - x_j: total output of sector j.
  - a_{ij}: units of good i required as an intermediate input to produce one unit of good j.
  - f_j: final demand for good j (consumers, government, exports).
- **Balance equation (sector j):**

$$x_j = \sum_i a_{ij} x_j + f_j$$

- In vector notation:

$$x = Ax + f$$

where A is the matrix of technical coefficients.

Productivity Condition in Input–Output Models

For the Leontief model

$$x = Ax + f \quad \Rightarrow \quad (I - A)x = f,$$

the matrix $(I - A)$ must be invertible for a unique solution x to exist.
This requires that the **technical coefficient matrix A be productive**.

- For each sector j, the column sum

$$\sum_i a_{ij} < 1$$

  means that producing one unit of output j uses **less than one unit of total inputs**.
- Leaves some output available for **final demand** ($f_j > 0$).
- If $\sum_i a_{ij} = 1$, the industry just produces to sell to other industries.
- If $\sum_i a_{ij} > 1$, the industry is not productive - uses more resources than it produces.

More formally:
The **spectral radius** (largest eigenvalue) of A satisfies $\rho(A) < 1$.

This guarantees $(I - A)^{-1}$ exists and the Leontief inverse is well-defined.

6.3 The Leontief Model

- Rearrange the balance equation:

$$x - Ax = f \quad \Rightarrow \quad (I - A)x = f$$

- Provided $(I - A)$ is invertible:

$$x = (I - A)^{-1} f$$

- The matrix $(I - A)^{-1}$ is called the **Leontief inverse**.

- Interpretation: it captures both the **direct** and **indirect** requirements needed to meet final demand f.

- Example: An increase in demand for cars requires more steel, which in turn requires more mining, etc.

6.4 Worked Example (3-sector economy)

Suppose three sectors: Agriculture (Ag), Manufacturing (M), and Services (S).

- **Technical coefficient matrix** the fraction of output from each sector used as input by others:
  - Rows = input sectors (what is used).
  - Columns = output sectors (what is produced).
  - Entry a_{ij} = units of input i needed per unit of output j. Ag needs 30% of its own output, 20% from M, 10% from S, etc.

$$A = \begin{bmatrix} 0.3 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ 0.2 & 0.1 & 0.3 \end{bmatrix}$$

Input / Output	Agriculture	Manufacturing	Services
Agriculture	0.3	0.2	0.1
Manufacturing	0.1	0.4	0.2
Services	0.2	0.1	0.3

- Final demand vector:

$$f = \begin{bmatrix} 100 \\ 150 \\ 200 \end{bmatrix}$$

- Solve:

$$x = (I - A)^{-1} f$$

- Interpretation: The required outputs x will be larger than f because each sector's output must also supply intermediates to others.

```r
# Technical coefficients (3×3)
A <- matrix(c(0.3,0.2,0.1,
              0.1,0.4,0.2,
              0.2,0.1,0.3), nrow = 3, byrow = TRUE)

I3 <- diag(3)

# Leontief inverse (you can also pre-specify L as below)
L <- solve(I3 - A)

# Base final demand
f <- matrix(c(100,150,200), ncol = 1)

# Compute gross outputs
x <- L %*% f
round(x, 5)
```

```
         [,1]
[1,] 336.7347
[2,] 455.1020
[3,] 446.9388
```

6.5 The Leontief Inverse and Multipliers

The matrix $(I - A)^{-1}$ is called the **Leontief inverse**.
Each element of this matrix has an important interpretation: it measures the **total requirement** of one sector's output needed to deliver one unit of final demand in another sector.

- The entry (i, j) of $(I - A)^{-1}$ tells us: "How many units of sector i are needed, directly and indirectly, to produce one additional unit of final demand for sector j."
- **Direct requirements:** given by A.
  - Example: if $a_{12} = 0.2$, producing 1 unit of sector 2 requires 0.2 units of sector 1 directly.
- **Indirect requirements:** captured through higher powers of A.
  - A^2 shows inputs needed two steps back (e.g., steel needed to make machine tools, which then make cars).
  - A^3 shows three-step linkages, and so on.
- The series expansion makes this clear (Nuemann):

  $$(I - A)^{-1} = I + A + A^2 + A^3 + \cdots$$

  So the Leontief inverse accumulates all **direct and indirect linkages**.

6.5.1 Multipliers

- A **multiplier** measures the total effect on the economy of a 1-unit increase in final demand for a given sector.
- In practice, the multiplier for sector j is the **column sum** of the jth column of $(I - A)^{-1}$.
  - It tells us how much total output across all sectors must rise to support 1 more unit of f_j.

- Sector-specific multipliers can also be read row by row:
  - The i, j entry shows the effect on sector i of a unit shock to demand in sector j.

6.5.2 Example (2-sector economy)

Suppose

$$A = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.2 \end{bmatrix}, \quad (I - A)^{-1} = \begin{bmatrix} 1.27 & 0.16 \\ 0.48 & 1.24 \end{bmatrix}.$$

- A 1-unit increase in demand for sector 1 requires 1.27 units of sector 1 output and 0.48 units of sector 2 output in total.

- Column sums: $1.27 + 0.48 = 1.75$.
  - Interpretation: a 1-unit increase in final demand for sector 1 leads to 1.75 total units of output economy-wide (a multiplier of 1.75).

- For sector 2, the multiplier is $0.16 + 1.24 = 1.40$.

Takeaway: The Leontief inverse is not just an algebraic trick — it directly encodes the multipliers that make input–output analysis such a powerful tool for tracing the effects of shocks.

6.6 Assumptions of the Input–Output Model

- **Fixed input coefficients (Leontief technology)**
  - Each sector uses inputs in fixed proportions.
  - Example: if it takes 0.3 units of steel to make a car, every car always uses 0.3 units — no substitution is allowed.
- **Constant returns to scale**
  - Doubling output requires exactly double the inputs.
  - There are no economies or diseconomies of scale.
- **Linearity**
  - Input requirements are linear in output.
  - The balance equation $x = Ax + f$ is valid for all production levels.
- **Homogeneous outputs**
  - Each sector produces a single, uniform good.
  - All uses of that good (whether as intermediate or final demand) are treated as identical.
- **No supply constraints**
  - Any level of final demand f can, in principle, be met — resources are not limited.
  - The model solves for required gross outputs without considering feasibility of factors like labor, land, or capital.
- **Static technology and demand**
  - The input coefficients A are assumed constant over time (no innovation or efficiency change within the model run).

– The model is comparative static — it compares equilibria before and after shocks, not dynamic adjustment paths.

- **Closed vs. open model treatment**
  – In the "open" model, households are part of final demand.

  – In the "closed" model, households are included as a sector that both consumes and supplies labor.

6.7 Input–Output Analysis in Practice

The input–output (I–O) model is usually used in a **calibration + shock** framework.
Rather than solving an optimization problem, we work with observed data to trace how the economy responds to changes in demand or technology.

1. Calibration (Base Year)

- Construct the **technical coefficient matrix** A from observed data:

$$a_{ij} = \frac{\text{input of } i \text{ used by } j}{\text{total output of } j}$$

  – Data typically come from national input–output tables.

- Observe base-year **final demand** f (household consumption, government, exports).

- Solve the identity

$$x = Ax + f \quad \Rightarrow \quad x = (I - A)^{-1} f$$

  to check consistency with observed outputs x.

- The calibrated model reproduces the base year exactly.

2. Apply a Shock

- Change final demand f:
  – Example: +10 units of exports of manufacturing.

- Or change technical coefficients A:
  – Example: new technology reduces electricity needed in agriculture.

- Or both.

3. Recompute Outputs

- Solve again:

$$x' = (I - A)^{-1} f'$$

  where f' is the new final demand (and possibly new A).

- The difference $x' - x$ shows the **direct and indirect effects** of the shock.

4. Interpret Results

- Which sectors expand the most?

- What are the **multipliers** (change in total output per unit change in final demand)?

- How do shocks propagate across the economy?

Example (illustrative):

- Suppose $f = (100, 150, 200)^\top$ and manufacturing exports rise by 20. And,

$$(I-A)^{-1} = \begin{bmatrix} 1.6 & 0.6 & 0.4 \\ 0.4 & 1.9 & 0.6 \\ 0..5 & 0.4 & 1.6 \end{bmatrix}$$

- Recalculate x'.
- Agriculture and services expand too, even though their final demand did not change — because they supply inputs to manufacturing.

```r
# Shock: +20 to Manufacturing final demand
df <- matrix(c(0,20,0), ncol = 1)
f_prime <- f + df

# New outputs after shock
x_prime <- L %*% f_prime

# Change in outputs
dx <- x_prime - x

list(
  x = round(x, 5),
  x_prime = round(x_prime, 5),
  delta_x = round(dx, 5)
)
```

```
$x
         [,1]
[1,] 336.7347
[2,] 455.1020
[3,] 446.9388

$x_prime
         [,1]
[1,] 348.9796
[2,] 493.4694
[3,] 455.9184

$delta_x
         [,1]
[1,] 12.24490
[2,] 38.36735
[3,]  8.97959
```

```r
# Manufacturing column of L (column 2)
L_col_M <- L[,2, drop=FALSE]
check_delta <- 20 * L_col_M

cbind(
  `20 * L[,2]` = round(check_delta, 5),
  `delta x`    = round(dx, 5)
```

```
)
```

```
          [,1]     [,2]
[1,] 12.24490 12.24490
[2,] 38.36735 38.36735
[3,]  8.97959  8.97959
```

This "calibrate + shock" procedure is the standard way input–output analysis is used in policy and research.

6.8 Relation to LP

We can recast the input–output system as a linear program. This allows us to apply the LP tools we've already studied (primal, dual, shadow prices) to questions of production and pricing.

- **Primal problem (quantities).**
  Decision variables are sector outputs x_j. The planner's problem might be to:
  - *Minimize cost* of meeting a fixed vector of final demands f, subject to input–output balance, or
  - *Maximize welfare* given resource or technology constraints.

  The balance condition $x = Ax + f$ can be rearranged as

  $$Ax + f \leq x,$$

  meaning that each sector's gross output must be at least enough to cover its intermediate input requirements (Ax) plus final demand (f).

- **Objective.**
  A typical choice is to minimize the total value of inputs required to satisfy demand. For example:

  $$\min\ c^\top x$$

  where c is a vector of unit costs. Alternatively, if outputs are valued in welfare terms, the objective can be to maximize $v^\top f$.

- **Constraints.**
  The inequalities $Ax + f \leq x$ ensure feasibility: sectors cannot promise more intermediate and final goods than they can produce.

- **Dual problem (prices).**
  The dual associates a variable p_i with each balance constraint. These p_i can be interpreted as **commodity prices** consistent with general equilibrium:
  - If a commodity is scarce (constraint binding), its price is positive.
  - If there is surplus, its shadow price falls to zero.

- **Strong duality.**
  At the optimum,

  $$\text{Value of outputs} = \text{Value of inputs}.$$

  In other words, the total revenue from selling goods at equilibrium prices equals the total cost of producing them using intermediate and primary inputs. This is the familiar condition for a competitive equilibrium in input–output economics.

Draw the parallels:

- *Primal* = "quantities world": how much each sector produces to satisfy demands.

- *Dual* = "prices world": what set of commodity prices makes those quantities consistent with equilibrium.

- Duality ensures these two views are perfectly consistent.

6.9 Applications in Economics

- **Environmental / Resource Economics**
  - Trace carbon, water, or energy embodied in consumption.
  - Assess land-use change from shifts in demand.
- **Policy Analysis**
  - Regional impact studies (IMPLAN, RIMS II).
  - Tariff or trade shock analysis.
- **Research extensions**
  - Hybrid I–O and LP models for energy systems planning.
  - Coupling I–O with CGE (computable general equilibrium) for price and income effects.

6.10 Classroom Activity

- Provide students with a small 2×2 A matrix and demand vector f.

- Ask them to compute:
  1. $(I - A)$
  2. $(I - A)^{-1}$
  3. $x = (I - A)^{-1} f$
- Discussion prompts:
  - Which sector has the larger multiplier?
  - What happens to required outputs if one coefficient a_{ij} increases (e.g., manufacturing requires more agriculture)?